

How to read this table.

- Asked quotes the reviewer's comment in full.
- Action taken is numbered in every row: 1 is what changed in the manuscript, 2 is why this move maximizes one-shot odds, 3 is the alternative we deliberately did not take and why it would lower odds.
- Remaining is color-coded: green Nothing means no further action, amber means an owner or template action is still needed, red means blocked on the PDF rebuild.

12 comments, one-shot resubmission. Manuscript sources: paper-access.tex (A) and paper-build.tex (B), kept in sync. Work line: main.

All comments in one table

Comment	Asked	Action taken	Remaining
R1-1 single collection, 31 images	Single dataset source and an extremely small manually verified benchmark: All experiments are carried out on the CGHD-1152 dataset only. The self-built human-verified benchmark merely contains 31 images with wide confidence intervals and insufficient coverage of sample distributions, which cannot prove that the proposed method can be universally adapted to various hand-drawing styles.	1. Action taken: results labeled as within-collection only. Drafter groups presented descriptively (31/31 assignments directly matched to unique drafter_N/images/ paths in the published CGHD-1152 v14 index, per-drafter table included): different hands give some style variation, but same corpus, so no universal-adaptation claim and no held-out-drafter experiment. Residual label bias disclosed as mitigated but not eliminated, with independently authored synthetic data as the second leg; future expansion described as future work. 34-annotated to 31-scored construction disclosed with exclusion reasons; join micro-F1 0.890 with the 95 percent bootstrap interval stated outright. 2. Why: N=31 cannot support a universal claim, and pretending otherwise fails verification. Honest bounding plus the synthetic second leg preserves credibility. 3. Alternative rejected: rushed expansion to 100-plus images, or a held-out-drafter experiment on tiny cells. Would produce noisy labels, risk missing the freeze, and read as tuned to the test set. Low-quality new data lowers odds more than clean disclosure.	Owner read-through. Drafter provenance closed: 31/31 direct filename matches to the published CGHD-1152 v14 index (v13 image paths agree); see docs/research/experiments/revision_evidence/drafter_map_n31_provenance.json. RE-VERIFY separately: 34-annotated to 31-scored accounting (Bosco).
R1-2 complex parts	The algorithm only supports basic passive and active discrete components: Complex components such as transformers, thyristors and optocouplers can only be located, while their pin connection relationships cannot be extracted, limiting the range of component types supported by netlist generation.	1. Action taken: new capability table separating three things the old text blurred together: • tested connectivity • estimated pins • illustrative export plus a metric-boundary row. Benchmark scope (R/C/L/D/Q plus voltage-source and IC groups) stated without validating universal device templates; named transformer type two-pin geometric guess, optocoupler templates, and unknown-type fallback pins disclosed without validated electrical pin semantics. Switch behavior stated exactly as the code does it: shorter OBB-edge midpoint pins with long-axis AABB fallback, fixed 0.001-ohm emission only if pins 0 and 1 map to distinct nodes, otherwise no switch line, no state or control model; ICs and listed complex devices with no export model skipped. Output and evaluation scope clarified in abstract, pipeline overview, metric definition, and conclusion. 2. Why: the detector merges original classes, so templates cannot establish separately detected device identities, and there is no ground truth to score complex-device pins. Exact code-truth wording replaces a blanket exclusion with something checkable. 3. Alternative rejected: implement full pin extraction for transformers, thyristors, and optocouplers in this revision. Unverifiable new capability with no evaluation data increases false-claim risk and would be caught on verification.	Owner read-through.

Comment	Asked	Action taken	Remaining
R1-3 hand-set thresholds	Reliance on manual tuning of fixed scale-relative thresholds: Hyperparameters including τ_{join} , τ_{t} and α need to be manually configured according to the median diagonal length of circuit components. Without an adaptive parameter learning mechanism, the detection accuracy tends to decline when processing circuits of extreme sizes.	1. Action taken: all values published plainly: • multipliers 0.62/0.30/0.20 times component scale, clamped to 24-60/11-28/8-20 pixels • alpha 0.35 fallback-only directional scoring with formula, used only when no component is assigned • completion reach 4x the clamped pin scale Existing reach-factor sweep 3 to 5 yielding macro-F1 0.895-0.903 reported; fixed-pixel base 0.820 versus scale-relative 0.816 kept as observed; method descriptions rewritten from the implementation; extreme sizes and controlled rescaling admitted as untested. 2. Why: adaptive parameter learning is a new research contribution, not a revision task, and cannot be validated on N=31 without looking tuned to the test set. Publishing the numbers plus the existing sweep bounds the claim with data already in hand. 3. Alternative rejected: build an auto-tuner or claim robustness to extremes. A rushed learner risks inconsistency with the frozen 192/192 numeric gate and invites questions it cannot answer.	Nothing.
R1-4 no values, cannot simulate	Lack of a closed-loop framework for component value recognition: The pipeline only reconstructs topological connections of circuits without integrating OCR to identify parameters of resistors, capacitors and voltage sources, so it cannot directly export complete simulation netlists that can be imported into SPICE.	1. Action taken: scope stated in abstract, intro, and conclusion; retitled to Structural Circuit Netlists including running heads and pipeline diagram node text; export labeled illustrative throughout, with external values and device models required and real-scan simulation equivalence unvalidated; related-work LTspice-validation implication on real scans removed; export substitutions and unsupported primitives disclosed. 2. Why: image-derived values and device models are outside the evaluated task. Keeping a simulatable-SPICE claim while reading no values is a false claim and an editor-level defect. 3. Alternative rejected: rush an OCR-plus-values loop to earn the old title. Low-quality OCR adds a new error source with no validation, and expands scope instead of fixing the discrepancy.	Nothing.
R1-5 small circuits only	Inability to handle complex circuits with multi-layer crossings and dense buses: The test samples adopted in this paper are dominated by simple topologies with 3 to 6 components. No dedicated verification is conducted for circuits with high-density multi-track buses and multi-layer crossed wiring, leading to a significant drop in recall on complex circuits.	1. Action taken: limits stated; true distribution printed in the caption (3-14 components per image in the evaluated electrical subset, median 7; 15 images at most five, 12 at least ten); dense buses and multilayer crossings stated as untested with nothing claimed beyond the data; crossover discussion narrowed; histogram asset preserved. 2. Why: the actual counts refute the 3-6-only premise, but count breadth does not validate buses or multilayer crossings. Descriptive counts plus an explicit boundary close the point honestly. 3. Alternative rejected: claim dense-bus support, or run an unplanned dense-bus evaluation with no data. No ground truth exists for that structure in the benchmark.	Nothing.

Comment	Asked	Action taken	Remaining
R1-6 memristor citations	It is suggested that the authors cite two papers in the sections related to SPICE simulation of memristive analog circuits: "A Memristor-Based Neural Network Circuit with Classical Conditioning and Fear Generalization" and "Biologically Plausible Memristive Decision-Making Circuit for Adaptive Control in Industrial Autonomous Navigation". Both papers complete full SPICE netlist modeling and simulation verification for memristive circuits, which complement the EDA technical route of converting hand-drawn schematics to SPICE netlists proposed in this paper in application scenarios, and can enrich relevant literature support for digital parsing and simulation deployment of analog memristive circuits.	1. Action taken: both papers checked at abstract level and cited in Related Work with full references (Gao et al., IEEE Trans. Consumer Electronics, Feb 2026; Gao et al., IEEE Trans. Industrial Informatics, vol. 22, 2026) as neighboring SPICE-deployment domains; no comparison or benchmark claim altered. 2. Why: straightforward compliance at near-zero cost. 3. Alternative rejected: argue relevance decline to keep the related work tight. Contesting a citation suggestion wastes goodwill for zero gain in a one-shot review.	Template check.
R2-1 statistics, roadmap	The scale of the human-verified benchmark is insufficient, casting doubt on the statistical reliability of the conclusions. The evaluation in this paper covers only 31 images. The authors should explicitly clarify the annotation cost and provide a roadmap for dataset expansion, or alternatively supply additional support through more rigorous statistical testing. If 31 images represent the entirety of currently available data, the wording of the conclusions should be made considerably more conservative.	1. Action taken: sampling in four strata with independent annotation and adjudication described; expansion protocol added as future work (stratified by component count, drafter, crossing/bus structure, and capture conditions, with per-image effort and provenance logging); missing timing honestly noted (single annotator, no per-image timing log, so no cost figure invented and no sample-size commitments); conclusions limited to 31 images from one corpus with bootstrap intervals retained; a CI containing zero stated as nonsignificance, not equivalence. 2. Why: conservative conclusions on the available data are exactly the fallback the reviewer offered. 3. Alternative rejected: invent per-image timing or promise a completed expansion. Fabricated cost data and false completion claims are the fastest way to lose editor trust.	Owner read-through.
R2-2 VLM comparison	There is a fundamental fairness issue in the VLM comparison experiment. The VLM enjoys an "oracle advantage" with respect to critical prior information, whereas the geometric pipeline must bear the burden of detection errors. In the paper, the VLM is provided with ground-truth annotated component bounding boxes; under such conditions, the fact that the VLM still shows no statistically significant difference from the geometric method does not sufficiently support the conclusion that "the geometric method is superior in terms of cost and structural validity."	1. Action taken: relabeled as an oracle-component-box connectivity diagnostic with identical annotated inputs to both methods (geometric method uses detected wires); VLM micro-F1 0.923 versus 0.890 with exact scores on 21/31 reported; paired difference +0.033 with 95 percent CI [-0.009, +0.078], inclusion of zero stated as nonsignificance rather than equivalence. Removed from the manuscript: • cost multiples and superiority claims • free-form-output criticism • autonomous bottleneck claims • global structural guarantees • the provisional autonomous paragraph (unauditable provenance) VLM section, intro comparison, abstract, real-join caption, discussion, and conclusion revised. 2. Why: the reviewer correctly identified the asymmetry. Keeping any superiority or cost claim under oracle conditions guarantees rejection. 3. Alternative rejected: keep the superiority framing, or run a new end-to-end VLM comparison with unauditable provenance. Both preserve the exact sentence the reviewer flagged.	Nothing.

Comment	Asked	Action taken	Remaining
R2-3 title overclaims	The paper title claims to generate SPICE netlists, yet the paper explicitly acknowledges that it "makes no attempt to read component values." This implies that the current system outputs a topological netlist rather than a simulatable SPICE netlist, which constitutes a notable discrepancy with the claims made in both the title and the abstract.	1. Action taken: retitled everywhere to Structural Circuit Netlists, including both running-head arguments in both manuscript sources and the pipeline diagram node text; metric boundary added across abstract, intro, pipeline overview, metric definition, capability table, and conclusion (component-pair metric does not certify pins, net partition, shorts, or simulation equivalence). 2. Why: title-body mismatch is the defect editors check first. Fixing it everywhere removes the objection instead of arguing it. 3. Alternative rejected: keep the SPICE title as conventional shorthand. That concedes the factual premise and would certainly fail.	Nothing.
R2-4 crossing wires	The impact of low recall in a critical component category on netlist correctness has not been sufficiently analyzed. The recall of the crossover class in component detection is only 70.7%, substantially lower than that of other categories. Misclassification of crossovers will introduce systematic errors in electrical connectivity, causing two wire segments that should remain electrically independent to be erroneously merged into the same net.	1. Action taken: full two-edge tradeoff reported (C242 suppressing two crossing-adjacent endpoint-endpoint edges: 27/4/0 to 19/0/8 — removing four false pairs loses eight true pairs, so no mitigation gain claimed); audit covers 13 annotated crossovers across eight images with the stored artifact kept; single-edge, rarity, and safe-suppression interpretations replaced; detector-miss effect originally retained as unmeasured because the join-isolation evaluation uses annotated boxes; it has since been measured conditionally: running the trained detector over the 31 benchmark images localized and classified all 13 annotated crossovers to the crossover class at confidence ≥ 0.25 (0.28-0.97; IoU 0.43-0.91), 12/13 at the 0.50 threshold, and a paired counterfactual deleting the annotated crossover boxes changed component-pair micro-F1 by +0.010 with identical net partitions on 23/31 images. Both are annotated-box, within-benchmark diagnostics, not held-out estimates (training-split membership unknown) and not an end-to-end detector-miss error rate. 2. Why: suppression is not a mitigation, and attributing observed false pairs to detector recall would be causal overreach. A conditional joining analysis plus a conditional detector check is what the design can measure. 3. Alternative rejected: retrain the detector to lift crossover recall, or claim the audit resolves the detector pathway. Retraining risks cross-class regressions before the freeze; held-out detector causality remains unmeasured by construction.	Nothing provable.

Comment	Asked	Action taken	Remaining
R2-5 ablations	The ablation study is severely insufficient.	1. Action taken: real table as observed: • base-graph/full-pipeline interventions for T-junctions, rail taps, directional preference, and base tolerance scaling • fixed-pixel base 0.820 versus scale-relative 0.816 and full-pipeline ties at 0.890 (418/37/66) presented as-is without spin • ties stated as neither identical connections nor proof of dispensability Activation and identical-connection explanations removed; method description rewritten (graph selection, mandatory slot/dummy assignment, guarded application). The two previously missing mechanism tests have since been run on the frozen 31-image benchmark and added as Table <code>tab:mec_ablation</code> / Section IV-E: occlusion removal drops component-pair micro-F1 from 0.890 to 0.336 (103/26/381; pairs change on 31/31 images), and the completion safeguards show the self-loop guard firing on 18/31 images rejecting 38 candidate merges with guard-off leaving scored pairs unchanged, plus witness-required at 0.884/0.907 (6/31 images changed). Run in detached worktrees; the frozen pipeline was not re-engineered, existing numbers are unchanged, and both arms were replayed independently to exact agreement. Removal of completion-wide scale dependence, and extreme/mixed-size inputs, remain untested and disclosed. 2. Why: keeping the unfavorable base result and the ties as an explicit negative result buys credibility, and the two requested mechanism tests convert a "not tested" disclosure into measured evidence. Adding results does not edit any existing number, so the numeric gate is untouched. 3. Alternative rejected: drop the fixed-pixel row, or re-engineer the pipeline pre-freeze to remove completion-wide scale dependence. Re-engineering the frozen pipeline would invalidate the evidence chain; a detached ablation of the shipped path does not.	Nothing provable.
R2-6 single-scalar limit	The setting of $\tau = k \cdot s$ may fail when component sizes within a single schematic vary substantially, and no relevant discussion addressing this limitation is provided.	1. Action taken: limitation stated and retained as asked — one scalar per image does not model intra-image size variation (large IC mixed with small discretes); clamps bound tolerances without adapting to heterogeneous symbols; fixed-pixel ablation changes the base graph only while completion retains clamped scale dependence; no rescaling experiment claimed; method text and sweep/ablation interpretation scoped accordingly, label anchor added. 2. Why: the reviewer asked for discussion, and discussion is full compliance at zero risk. 3. Alternative rejected: rebuild tau as per-component adaptive in this revision. That is a new method requiring new validation, with no ground truth designed to score it.	Nothing.

Shared language everywhere: a non-finding is not equivalence; no blanket guarantees; pair scores do not certify pin layouts; lab-condition results do not prove real-world dominance.

Reference numbers

Join **0.890 (418/37/66)** on all 31; perfect-wire **0.88983 (420/40/64)**. Wire ledger: **0.9755** equals the 10-degree/18px config, **0.9726** equals the 12-degree/8px production. Otsu **0.789**. VLM **0.923, 21/31** exact. Reach sweep macro **0.895-0.903**. Numeric gate: **192/192** stored comparisons pass.